Zain Usmani

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Education

University of Windsor 2024 - Present
BASc in Electrical and Computer Engineering (Minor in CS) GPA: 3.7/4.0

Experience

Embedded Systems Developer June 2025 – Present
Formula Electric - Lancer Motorsport

- Analyzed signal integrity and debugged physical hardware buses using oscilloscopes and logic analyzers
- Designed a high-performance dashboard on RPi OS Linux, bridging C++ and Linux backend services and vehicle parameters with a QT graphical interface in real-time
- Built automated Python scripts to validate embedded system telemetry and data transmission integrity

FPGA Vision Intern Apr 2026 – Aug 2026
O-Hive

- Engineered a real-time computer vision pipeline in C++ and deployed it to an AMD Kria KV260 embedded Linux system using a custom CMake build architecture
- Quantized FastDepth weights to 8 bit integer precision using Brevitas and QONNX
- Containerized firmware workspace via Docker to manage dependencies and ensure cross-compilation
- Managed Linux V4L2 drivers and Xilinx firmware to route raw video directly to hardware accelerators
- Designed a C++ memory fork to split and route incoming NV12 video buffers to FPGA accelerator
- Corrected metric distortion by designing a algorithm using physical focal lengths to fix 4.6x error margin

ML and Data Intern Jan 2025 – Sep 2025
MicroNano Mechatronics Lab

- Programmed an ESP32 Bluetooth Low Energy pipeline in C++ to stream live IMU data at 50 Hz
- Executed analysis on 16000 sensor samples using Pandas and NumPy to extract feature vectors
- Trained XGBoost classification models to distinguish human gait patterns achieving a 78.9 F1 score
- Co-authoring research paper detailing the machine learning performance for biomechanical classification

Student Campus Ambassador Dec 2024 – June 2026
UWindsor Recruitment

- Guided weekly facility tours for prospective engineering students to showcase technical laboratories
- Represented the engineering faculty at recruitment events such as Open House and Grade 12 Day

Projects

RouteScout Rover (McHacks 13) | *Embedded C++, RPi 5, ESP32, 3D CAD* Jan 2026

- Developed a real-time odometry pipeline on ESP32, which relays data back to RPi computer
- Captured high frequency data from motor encoders and IMU registers via hardware interrupts with I2C
- Implemented a Madgwick filter and closed-loop PID control to generate precise PWM motor actuation

VGA Timing Controller and Frame Buffer | *Verilog, Intel Quartus, MAX 10 FPGA* Apr 2026

- Designed a hardware VGA controller in Verilog to drive a 640x480 pixel display on a MAX 10 FPGA
- Designed dual-port SRAM buffer to decouple video timing requirements from pixel generation logic
- Managed memory read timing constraints and synthesized RTL design to achieve target clock frequencies

Skills

Languages and Tools: C++, C, Python, Linux (Ubuntu on SoM), Shell Scripting, CMake/Ninja, Git
Hardware: Oscilloscope, Waveform Generator, Logic Analyzer, Multimeters, SystemVerilog, Quartus
Embedded Systems: AMD Kria KV260, STM32, ESP32, Raspberry Pi, I2C, SPI, USART, Cube IDE
Concepts: Microarchitecture, Signal Integrity, I2C and SPI Protocols, DMA, PID Control, Sensor Fusion